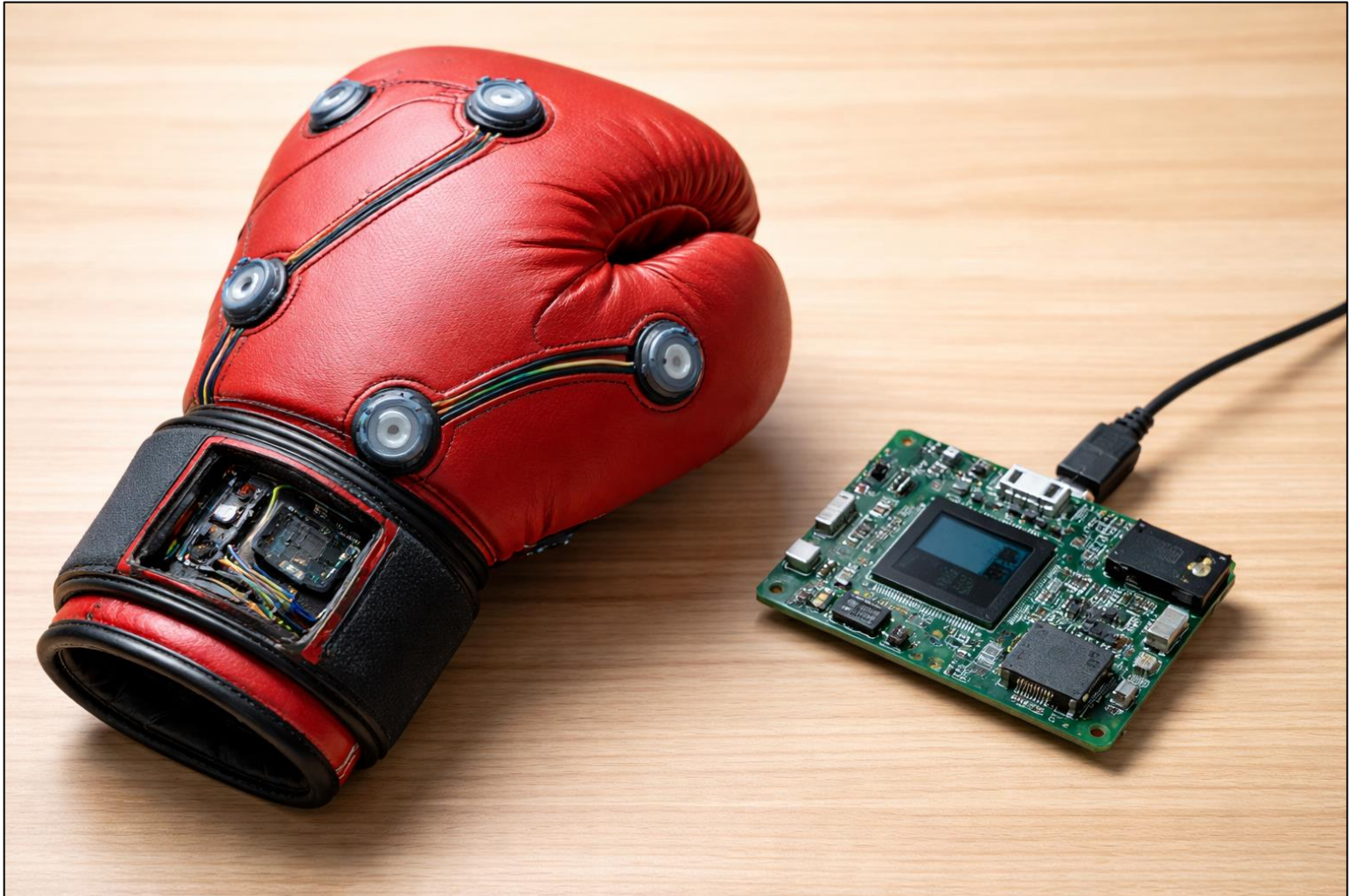




SMART BOXING GLOVE

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OBJECTIVE

The project aims to develop a wearable smart boxing glove system. The primary goal is to quantify athletic performance by measuring impact force, velocity, and 3D motion. This involves designing a low-power embedded system based on the ESP32-C3 to capture and transmit high-fidelity sensor data. The final solution will provide real-time feedback via a computer interface, merging electronics with sports science to create a practical training tool validated by professional supervision.

MOTIVATION

We chose this project due to our shared interest in boxing and our understanding that many people train by themselves at home or even in boxing gyms without personalized feedback. In many training environments, especially outside professional gyms, measurements of punch strength, speed and motion patterns are not available. With feedback you can see right away changes in how training feels, progress becomes visible, small shifts in technique stand out. This project represents an opportunity to combine sports and engineering in a meaningful way, applying technology to solve a practical problem in athletic performance analysis.



PROJECT DESCRIPTION

The system consists of a sensor-instrumented glove that maintains its ergonomics for safe use. The electronics (microcontroller and battery) will be housed in a protective casing (still to be decided where to be put, because it needs to be in a spot that avoids the impact area). It integrates a 9-DOF IMU, force sensor and accelerometer. The data will be processed and sent to a PC application responsible for real-time 3D trajectory reconstruction and impact analysis on an X, Y, Z coordinate system.

PROBLEMS AND APPROACHES

- **CONNECTIVITY:** we must choose between Bluetooth (high compatibility) and ESP-NOW (low latency). Our approach will be to prioritize the protocol that ensures the highest stability for real-time 3D rendering.
- **EVENT TRIGGERING:** a major challenge is distinguishing a punch from a defensive block. We plan to implement threshold-based algorithms to trigger data recording only when a specific acceleration pattern is detected.
- **MECHANICAL DURABILITY:** to protect the electronics from high impacts, we will design a 3d-printed enclosure with internal damping, ensuring the safety of both the user and the components.

KEY CONCEPTS

ESP-NOW: A fast, connectionless communication protocol by Espressif. It allows the glove to send data packets directly to the PC with minimal delay (peer to peer).

TASK SPLIT

For this project we are planning to divide into 2 groups of tasks:

- Student 1 (Hardware/Firmware): Electronic assembly, implementation of the communication protocol and power management.
- Student 2 (Software/Processing): Development of the PC application, real-time data visualization, 3D reconstruction of the punch and data logging.